

Safety data sheet

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BASF Safety data sheet
Date / Revised: 16.02.2024
Product: **Protectol® BN**

Version: 5.0

(30212030/SDS_GEN_TH/EN)

Date of print: 25.02.2026

1. Substance/preparation and manufacturer/supplier identification

Product name:
Protectol® BN

Use: biocide

Manufacturer/supplier:
BASF (Thai) Limited
23rd Floor, Emporium Tower, 622, Sukhumvit 24 Rd.,
Klongton, Klongtoey, Bangkok 10110, THAILAND
Telephone: +66 2624-1999
Telefax number: +66 2664-9254
E-mail address: Thailand-SDS-info@basf.com

Emergency information:
International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification according to UN GHS 2009

Classification of the substance and mixture:

Acute toxicity: Cat.3 (Inhalation - dust)

Acute toxicity: Cat.3 (oral)

Acute toxicity: Cat.4 (dermal)

Skin irritation: Cat.2

Serious eye damage: Cat.1

Specific target organ toxicity — single exposure (Respiratory system): Cat.3 (irritating to respiratory system)

Hazardous to the aquatic environment - acute: Cat.1

Hazardous to the aquatic environment - chronic: Cat.1

M-factor acute: 100

M-factor chronic: 10

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M-factor acute: 10
 M-factor chronic: 10

Label elements and precautionary statement:

Pictogram:



Signal Word:
 Danger

Hazard Statement:

H318	Causes serious eye damage.
H315	Causes skin irritation.
H312	Harmful in contact with skin.
H335	May cause respiratory irritation.
H301 + H331	Toxic if swallowed or if inhaled.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves, protective clothing and eye protection or face protection.
P273	Avoid release to the environment.
P261	Avoid breathing dust.
P261	Avoid breathing dust or fume.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P310	Immediately call a POISON CENTER or physician.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P330	Rinse mouth.
P361 + P364	Take off immediately all contaminated clothing and wash it before reuse.
P391	Collect spillage.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
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Other hazards which do not result in classification:

Fine dust produced by abrasion can form explosive mixtures with air.

3. Composition/information on ingredients

Chemical nature

Substance nature: Substance

bronopol (Content (W/W): > 99 %)
CAS Number: 52-51-7

4. First-Aid Measures

General advice:

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On skin contact:

Immediately wash thoroughly with plenty of water, apply sterile dressings, consult a skin specialist.

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
dry powder, foam

Unsuitable extinguishing media for safety reasons:
carbon dioxide

Additional information:

Avoid whirling up the material/product because of the danger of dust explosion.

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Specific hazards:

harmful vapours, halogenated compounds, hydrogen halides, carbon oxides, nitrogen oxides

Evolution of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:

Wear a self-contained breathing apparatus.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations. Dusty conditions may ignite explosively in the presence of an ignition source causing flash fire.

6. Accidental Release Measures

Personal precautions:

For non-emergency personnel: Avoid dust formation. Use personal protective clothing. Information regarding personal protective measures, see section 8.

For emergency responders: Take appropriate protective measures.

Environmental precautions:

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:

For small amounts: Contain with dust binding material and dispose of.

For large amounts: Pick up with suitable appliance and dispose of.

Avoid raising dust. Dispose of absorbed material in accordance with regulations.

Additional information: Avoid the formation and build-up of dust - danger of dust explosion. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition.

7. Handling and Storage

Handling

Provide exhaust ventilation.

No eating, drinking, smoking or tobacco use at the place of work. Wash hands before breaks and at end of work. Remove contaminated clothing and protective equipment before entering eating areas.

Protection against fire and explosion:

Avoid dust formation. The product is capable of dust explosion. Take precautionary measures against static discharges. Avoid all sources of ignition: heat, sparks, open flame. Avoid deposition of dust.

Storage

Suitable materials for containers: High density polyethylene (HDPE), Low density polyethylene (LDPE), Polypropylene (PP), glass, Polyethyleneterephthalate (PET), Aluminium
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

8. Exposure controls and personal protection

Components with occupational exposure limits

No occupational exposure limits known.

Personal protective equipment

Respiratory protection:

Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with high efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P3 or FFP3).

Hand protection:

Chemical resistant protective gloves

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):

e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Tightly fitting safety goggles (cage goggles) (e.g. EN 166) and face shield.

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Wearing of closed work clothing is required additionally to the stated personal protection equipment.

No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form:	crystalline
Colour:	white
Odour:	product specific
Odour threshold:	not determined

pH value:	5.7 (water, 1 %(m))	(DIN ISO 976)
pKA:	9.56 (21 °C)	(other)

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Melting point:	129 °C	(Directive 92/69/EEC, A.1)
Boiling point:	(1,013.25 hPa) The substance / product decomposes therefore not determined. The substance / product decomposes.	(other)
Flash point:	not applicable, the product is a solid	
Evaporation rate:	The product is a non-volatile solid.	
Flammability (solid/gas):	not highly flammable	(Directive 92/69/EEC, A.10)
Lower explosion limit:	For solids not relevant for classification and labelling.	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Ignition temperature:	not applicable	
Thermal decomposition:	155 °C , 2,870 kJ/kg	(DSC (OECD 113))
Self ignition:	not self-igniting	Test type: Spontaneous self-ignition at room-temperature.
	not self-igniting	Test type: Self-ignition at high temperatures. (Method: Directive 92/69/EEC, A.16)
Self heating ability:	It is not a substance capable of spontaneous heating according to UN transport regulations class 4.2.	
SADT:	> 75 °C Heat accumulation / Dewar 500 ml (SADT, UN-Test H.4, 28.4.4)	
Minimum ignition energy:	> 1 J (13 °C)	(VDI 2263, sheet 1, 2.5 (May 1990))
Explosion hazard:	not explosive	(Directive 92/69/EEC, A.14)
Fire promoting properties:	not fire-propagating	(Directive 92/69/EEC, A.17)
Vapour pressure:	0.000051 hPa (20 °C)	(Directive 92/69/EEC, A.4)
Density:	1.905 g/cm ³ (20 °C)	(OECD Guideline 109)
Relative density:	1.905 (20 °C)	(OECD Guideline 109)
Bulk density:	approx. 1,100 kg/m ³	
Relative vapour density (air):	The product is a non-volatile solid.	

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Solubility in water:

286 g/l
(20.2 °C)

Hygroscopy: The product has not been tested.

Partitioning coefficient n-octanol/water (log Pow): 0.22 (Directive 92/69/EEC, A.8)
(24 °C; pH value: 7)Surface tension: 72 mN/m (Directive 92/69/EEC, A.5,
(20 °C; 1 g/l) OECD harmonized ring
method)

Viscosity, dynamic:

not applicable

Viscosity, kinematic:

not applicable, the product is a solid

Molar mass: 199.99 g/mol

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Conditions to avoid:

Avoid all sources of ignition: heat, sparks, open flame. Avoid heat. Avoid dust formation. Avoid deposition of dust.

Thermal decomposition: 155 °C, 2,870 kJ/kg (DSC (OECD 113))

Substances to avoid:

oxidizing agents
sodium hydroxide, bases
aluminium, metal

Corrosion to metals: No corrosive effect on metal.

Hazardous reactions:

The product may contain explosive fine dust or such dust may be produced by abrasion during transport or product transfer.

Hazardous decomposition products:

nitrogen oxides, Bromine compounds

Chemical stability:

The product is stable if stored and handled as prescribed/indicated.

Reactivity:

No hazardous reactions if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:

LD50rat (oral): > 50 - < 300 mg/kg (OECD Guideline 401)

Acute inhalation toxicity

LC50 rat (by inhalation): > 0.5 - < 1 mg/l 4 h

An aerosol was tested.

Acute dermal toxicity

LD50 rat (dermal): > 1,000 - < 2,000 mg/kg (OECD Guideline 402)

Assessment of acute toxicity

Of moderate toxicity after short-term skin contact. Of moderate toxicity after single ingestion. Of moderate toxicity after short-term inhalation.

Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far

Irritation

Assessment of irritating effects:

Irritating to skin. May cause severe damage to the eyes.

Experimental/calculated data:

Skin corrosion/irritation rabbit: Irritant. (OECD Guideline 404)

Serious eye damage/irritation rabbit: irreversible damage (Draize test)

The European Union (EU) has classified this substance with 'Risk of serious damage to eyes'.

Respiratory/Skin sensitization

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Experimental/calculated data:

Guinea pig maximization test guinea pig: Non-sensitizing. (similar to OECD guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was mutagenic in a mammalian cell culture test system. No mutagenic effect was found in various tests with bacteria and mammals.

Carcinogenicity

Assessment of carcinogenicity:

The substance showed no carcinogenic activity in animals after chronic administration to the skin. In long-term animal studies in which the substance was given in the drinking water in high doses, a carcinogenic effect was not observed.

Reproductive toxicity

Assessment of reproduction toxicity:

The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:

No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Experiences in humans

Experimental/calculated data:

If this substance comes into close contact with the skin of hypersensitive persons, sensitization might occur.

Specific target organ toxicity (single exposure)

Causes temporary irritation of the respiratory tract.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

After repeated exposure the prominent effect is local irritation.

Aspiration hazard

not applicable

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

Depending on local conditions and existing concentrations, disturbances in the biodegradation process of activated sludge are possible. Very toxic to aquatic life.

Toxicity to fish:

LC50 (96 h) > 10 - < 100 mg/l, Fish (EPA 72-1)

Aquatic invertebrates:

EC50 (48 h) > 1 - < 10 mg/l, Daphnia magna (OECD Guideline 202, part 1)

Aquatic plants:

EC50 (72 h) < 1 mg/l (growth rate), Selenastrum capricornutum (OECD Guideline 201)
acute Effect

Chronic toxicity to fish:

No observed effect concentration (28 d) > 1 - < 10 mg/l, Oncorhynchus mykiss (OECD Guideline 215, Flow through.)

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (21 d), > 0.1 - < 1 mg/l, Daphnia magna (OECD Guideline 211, Flow through.)

The statement of the toxic effect relates to the analytically determined concentration.

Assessment of terrestrial toxicity:

Mobility

Assessment transport between environmental compartments:

The substance will not evaporate into the atmosphere from the water surface.

Adsorption to solid soil phase is not expected.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

Moderately/partially biodegradable.

Assessment of stability in water:

In contact with water the substance will hydrolyse rapidly.

Information on Stability in Water (Hydrolysis):

t_{1/2} 2.4 h (50 °C), (OECD Guideline 111, pH 7)

Bioaccumulation potential

Assessment bioaccumulation potential:

Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Additional information

Add. remarks environm. fate & pathway:

Treatment in biological waste water treatment plants has to be performed according to local and administrative regulations.

Other ecotoxicological advice:

Do not allow to enter soil, waterways or waste water channels.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.

No disposal via sewage or waste water systems.

Contaminated packaging:

Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

UN number or ID number: UN 3241

UN proper shipping name: 2-BROMO-2-NITROPROPANE-1,3-DIOL

Transport hazard class(es): 4.1, EHSM

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Packing group: III
Environmental hazards: yes

Special precautions for user: Protect from heat

Sea transport

IMDG

UN number or ID number: UN 3241
UN proper shipping name: 2-BROMO-2-NITROPROPANE-1,3-DIOL
Transport hazard class(es): 4.1, EHSM
Packing group: III
Environmental hazards: yes
Marine pollutant: YES
Special precautions for user: EmS: F-J; S-G
Protect from heat

Air transport

IATA/ICAO

UN number or ID number: UN 3241
UN proper shipping name: 2-BROMO-2-NITROPROPANE-1,3-DIOL
Transport hazard class(es): 4.1
Packing group: III
Environmental hazards: No Mark as dangerous for the environment is needed
Special precautions for user: Protect from heat

15. Regulatory Information

Other regulations

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Information on intended use: This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use. This includes the mentioned and recommended usage. In particular this concerns the application for products that are the object of special standards and regulations. Any other intended applications should be discussed with the manufacturer.

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Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

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